

Gene Expression Classification with Support Vector Machines

Statistics for Data Science — Homework #3

University of Trento

MSc in Data Science

Author: Henri Vasserot

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Objective

This report investigates the use of Support Vector Machine (SVM) classifiers for predicting leukemia subtypes from high-dimensional gene expression data. The analysis compares classification performance using the complete set of genes and a variance-filtered subset containing the top 5% most variable genes.

The workflow includes:

- exploratory data analysis,
- train/test evaluation,
- hyperparameter tuning with cross-validation,
- comparison of linear and RBF kernels,
- variance-based dimensionality reduction,
- discussion of overfitting and generalization performance.

The report is fully reproducible and the complete source code is available through the accompanying GitHub repository.

GitHub repository



Complete reproducible code and notebook:

https://github.com/lolipop913/homework03_svm_gene_expression.git

Gene Expression Classification with Support Vector Machines

This study investigates the use of support vector machine (SVM) classifiers for predicting leukemia subtypes from high-dimensional gene expression data. The dataset consists of expression measurements for 2000 genes collected from 79 patients belonging to two groups: patients with a chromosomal translocation and cytogenetically normal patients.

Gene expression data are characterized by a high-dimensional setting in which the number of predictors greatly exceeds the number of observations. Such settings are particularly prone to overfitting and require careful model selection and validation strategies. Support vector machines are well suited to this context because they can handle high-dimensional predictors while controlling model complexity through margin maximization and regularization.

The analysis follows a supervised learning framework including data exploration, hyperparameter tuning, model evaluation on unseen data, and comparison of different SVM variants. In addition, a variance-based gene filtering strategy is investigated in order to assess whether restricting the analysis to the most variable genes improves predictive performance and model stability.

1. Imports and setup

The analysis relies on standard Python libraries for data manipulation, visualization, model fitting, hyperparameter tuning, and performance evaluation.

2. Load data

The dataset contains gene expression measurements for 79 patients and 2000 genes, together with a binary response variable indicating the leukemia subtype. Before fitting the models, we inspect the dimensionality of the dataset and the class distribution in order to identify potential issues such as class imbalance and the high-dimensional nature of the predictors.

```
X shape: (79, 2000)
y shape: (79,)
Class distribution:
-1    42
 1    37
Name: count, dtype: int64
```

The dataset contains 79 observations and 2000 gene expression variables, leading to a high-dimensional setting in which the number of predictors greatly exceeds the number of samples. Such configurations increase the risk of overfitting and motivate the use of regularized classification methods such as support vector machines.

The two leukemia subgroups are relatively balanced (42 versus 37 observations), which reduces the risk of severe classification bias toward one class and allows accuracy to remain a meaningful evaluation metric.

3. Train/Test split

The dataset is divided into training and test sets in order to evaluate predictive performance on unseen observations. This separation is particularly important in high-dimensional settings, where complex models may easily overfit the training data and produce overly optimistic performance estimates.

A stratified split is used to preserve the class proportions in both subsets. Model selection and hyperparameter tuning are performed exclusively on the training set, while the test set is kept aside for the final evaluation of generalization performance.

```
Train size: (55, 2000)
Test size: (24, 2000)
Train class distribution:
-1    0.527273
 1    0.472727
Name: proportion, dtype: float64
Test class distribution:
-1    0.541667
 1    0.458333
Name: proportion, dtype: float64
```

The dataset is divided into 55 training observations and 24 test observations. The class proportions remain similar across the two subsets, confirming that the stratified sampling strategy successfully preserved the balance between leukemia subgroups.

Maintaining comparable class distributions between training and test sets is important to ensure that performance estimates are not distorted by differences in class composition across the data partitions.

4. Analysis plan

Two support vector machine classifiers are considered: a linear SVM and a radial basis function (RBF) SVM. The linear kernel provides a simple and interpretable decision boundary, while the RBF kernel allows for more flexible non-linear classification patterns. For both models, the regularization parameter (C) is tuned in order to control the trade-off between margin maximization and classification errors. For the RBF kernel, the parameter (γ) is additionally optimized, as it controls the flexibility of the decision boundary.

Model performance is assessed using a stratified train/test split together with cross-validation on the training set. Hyperparameter tuning is performed using `GridSearchCV`, and the final evaluation is carried out on the independent test set in order to obtain an unbiased estimate of predictive performance.

Two preprocessing strategies are compared. The first uses the complete set of genes, while the second restricts the analysis to the 5% most variable genes. Variance-based filtering is commonly used in gene expression studies to reduce noise and dimensionality, since genes with very low variability often contribute little discriminative information.

5. Data exploration

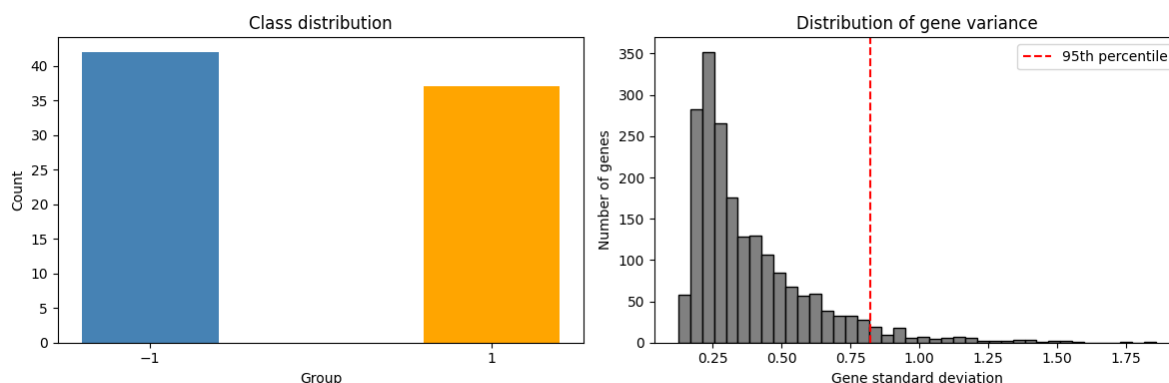
We first examine the class distribution and the variability of gene expression across samples. The two leukemia subgroups appear reasonably balanced, which reduces the risk of severe clas-

sification bias toward one class and supports the use of accuracy as a meaningful performance measure.

Many genes may contribute mainly noise rather than useful discriminative information, potentially reducing model stability and generalization performance.

To investigate this issue, we inspect the distribution of gene standard deviations across the dataset. The results show substantial heterogeneity in gene variability: while a small subset of genes exhibits high variability across patients, many genes display relatively low variation. Genes with very low variability are less likely to contribute to class discrimination and may unnecessarily increase the dimensionality of the feature space.

This motivates the use of variance-based filtering, where only the top 5% most variable genes are retained for downstream analysis. Since the filtering procedure is incorporated within the machine learning pipeline, feature selection is performed using only the training data in each resampling step, thereby avoiding information leakage.



Total number of genes: 2000

Top 5% genes: 100

The class distribution confirms that the two leukemia subgroups are relatively balanced, with 42 observations in class (-1) and 37 observations in class (1). This reduces the risk of strong classification bias toward one group and supports the use of accuracy as an informative evaluation metric.

The distribution of gene standard deviations reveals substantial heterogeneity in gene variability across the dataset. While most genes exhibit relatively low variability, a smaller subset displays markedly larger standard deviations. This pattern supports the use of variance-based filtering, since highly variable genes are more likely to contain discriminative information, whereas low-variance genes may contribute mainly noise.

Using the 95th percentile threshold retains 100 genes out of the original 2000 predictors. This substantially reduces the dimensionality of the feature space while preserving the most variable expression patterns for downstream classification.

6. Pipelines and utility functions

Two machine learning pipelines are defined: one using the complete set of genes and another incorporating variance-based gene filtering. In both cases, preprocessing and model fitting are integrated within a single pipeline structure in order to ensure that all transformations are applied consistently during cross-validation.

This approach is particularly important in high-dimensional settings, since feature selection performed outside the resampling procedure may introduce information leakage and lead to overly optimistic estimates of predictive performance.

The same cross-validation strategy is used across all models to ensure a fair comparison between preprocessing approaches and SVM variants.

7. SVM on all features

We first fit support vector machine classifiers using the complete set of available genes. This initial analysis provides a baseline for evaluating predictive performance in the original high-dimensional feature space, without any prior dimensionality reduction or feature filtering.

Both the linear and RBF kernels are evaluated after hyperparameter tuning through cross-validation on the training set. The comparison between kernels allows us to assess whether a more flexible non-linear decision boundary improves classification performance in this setting.

Because the number of predictors is substantially larger than the number of observations, this analysis is also expected to highlight the potential impact of high dimensionality on model complexity and generalization performance.

```
Full features - kernel: rbf
Full features - C: 10
Full features - CV accuracy: 0.691
Full features - train accuracy: 1.0
Full features - test accuracy: 0.667
```

For the full-feature analysis, the best-performing model selected by cross-validation is an RBF SVM with ($C = 10$) and ($=$) "scale". The selection of a relatively large (C) suggests that the model favors a more complex decision boundary with reduced tolerance for classification errors on the training data.

The model achieves a cross-validation accuracy of approximately 0.69 and a test accuracy of 0.67. In contrast, the training accuracy reaches 1.00, indicating perfect classification on the training set. The large discrepancy between training and test performance strongly suggests

overfitting, which is expected in a setting where the number of predictors substantially exceeds the number of observations.

This behavior illustrates the difficulty of learning stable decision boundaries in high-dimensional gene expression data. Although the flexible RBF kernel can capture complex patterns in the training data, part of this structure may correspond to noise rather than reproducible discriminative signal.

8. SVM with top 5% variance filtering

We next repeat the analysis after restricting the predictors to the 5% most variable genes. Variance-based filtering is commonly used in gene expression studies because many genes exhibit little variation across samples and are therefore less likely to contribute meaningful discriminative information.

Reducing the dimensionality of the predictor space may improve model stability and generalization performance by limiting the influence of noisy or weakly informative variables. This issue is particularly relevant in the present dataset, where the number of predictors greatly exceeds the number of observations.

The filtering procedure is integrated directly within the machine learning pipeline. Consequently, the subset of selected genes is determined independently within each resampling step using only the training data. This prevents information leakage and ensures that the estimated predictive performance remains unbiased.

The filtered analysis is then compared with the full-feature analysis in order to evaluate whether restricting the model to highly variable genes improves classification performance and model robustness.

```
Filtered features - kernel: rbf
Filtered features - C: 1
Filtered features - CV accuracy: 0.782
Filtered features - train accuracy: 0.982
Filtered features - test accuracy: 0.75
Number of selected features: 100
```

After variance-based filtering, the best-performing model remains an RBF SVM, but with a smaller regularization parameter ($C = 1$). Compared with the full-feature analysis, the filtered model achieves higher cross-validation accuracy (0.78 versus 0.69) and improved test accuracy (0.75 versus 0.67).

The train accuracy remains high (0.98), but the gap between training and test performance is smaller than in the full-feature setting. This suggests that restricting the analysis to the most variable genes reduces overfitting and improves generalization performance.

Only 100 genes are retained after filtering, corresponding to a substantial reduction in dimensionality relative to the original 2000 predictors. These results are consistent with the idea that many low-variance genes contribute mainly noise rather than useful discriminative information. Removing such variables may therefore help the model identify more stable and reproducible classification patterns.

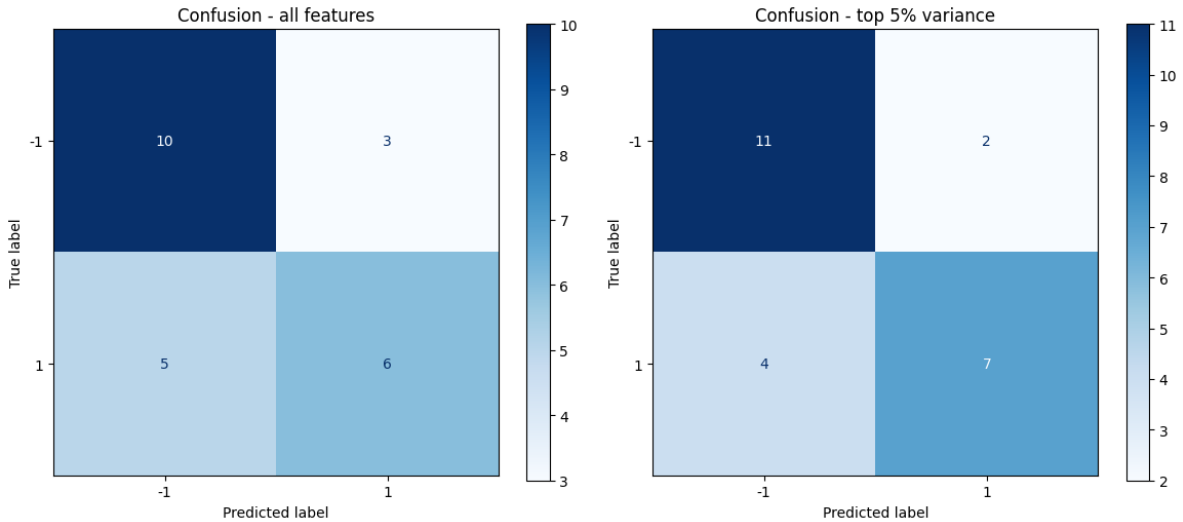
The next section directly compares the two preprocessing strategies in terms of predictive performance and generalization ability.

9. Performance comparison

The predictive performance of the different SVM models is compared using the same training and test sets in order to ensure a fair evaluation across preprocessing strategies and kernel choices. Using identical data partitions allows performance differences to be attributed primarily to the modeling approach rather than to random variations in the sample split.

The comparison focuses on the effect of variance-based filtering on classification accuracy and generalization performance. In particular, we assess whether restricting the analysis to highly variable genes improves predictive performance, reduces model complexity, and leads to more stable results in this high-dimensional setting.

	approach	cv_accuracy	train_accuracy	test_accuracy	diff
0	all features	0.691	1.000	0.667	0.333
1	top 5% variance	0.782	0.982	0.750	0.232



The comparative results indicate that variance-based filtering improves predictive performance across all evaluation stages. The filtered model achieves higher cross-validation accuracy (0.78

versus 0.69) and higher test accuracy (0.75 versus 0.67) than the model trained on the complete set of genes.

The best-performing model in both settings is an RBF SVM. However, the filtered analysis selects a smaller regularization parameter ($C = 1$) compared with the full-feature model ($C = 10$). This suggests that, after dimensionality reduction, a less complex decision boundary is sufficient to achieve better predictive performance.

The difference between training and test accuracy also decreases from 0.33 in the full-feature analysis to 0.23 after filtering. Although both models still exhibit some degree of overfitting, the smaller train-test gap suggests that the filtered model generalizes better to unseen observations.

The confusion matrices further support this interpretation. In the full-feature setting, the classifier produces several misclassifications in both classes, particularly for observations belonging to class (1). After variance filtering, the number of correctly classified observations increases for both groups, resulting in fewer overall classification errors and improved test accuracy.

Only 100 genes are retained after filtering, corresponding to a substantial reduction relative to the original 2000 predictors. These results suggest that restricting the analysis to highly variable genes improves the discriminative structure of the feature space and leads to more stable classification patterns.

10. Conclusion

This analysis investigated the use of support vector machine classifiers for leukemia subtype prediction using high-dimensional gene expression data. Both linear and RBF kernels were evaluated on the complete feature space and after variance-based dimensionality reduction.

Overall, the filtered approach produced more stable predictive performance than the full-feature model, suggesting that dimensionality reduction can be beneficial in high-dimensional gene expression settings.

More broadly, the analysis highlights the methodological challenges associated with high-dimensional datasets in which the number of predictors greatly exceeds the number of observations. In such settings, complex models may easily adapt to noise in the training data, making rigorous validation procedures and careful preprocessing particularly important.

Several limitations should nevertheless be acknowledged. The dataset remains relatively small, the variance threshold is heuristic, and no external validation dataset is available. Consequently, the reported performances should be interpreted cautiously and may not fully generalize to independent patient populations.

Overall, the results illustrate how dimensionality reduction and regularized machine learning methods can improve classification performance in high-dimensional gene expression studies while also emphasizing the importance of controlling overfitting.

Final summary

- Full features test accuracy: 0.667
- Filtered features test accuracy: 0.75
- Filtered model train-test gap: 0.232

The filtered model still exhibits some degree of overfitting, although the train-test gap is smaller than in the full-feature analysis.